PROBLEM SET 5, PART 1: LIE GROUPS AND LIE ALGEBRAS
DUE: NOV. 28

(1) [**Left invariant vector fields**]
Let X be a left invariant vector field on a Lie group G. Prove:
(a) X is smooth.
(b) X is complete.

(2) [**The exponential map for matrix Lie groups**]
Define the norm of a matrix $A = (a_{ij})$ to be $\|A\| = (\sum_{i,j} |a_{ij}|^2)^{1/2}$. It is easy to check $\|A + B\| \leq \|A\| + \|B\|$ and $\|AB\| \leq \|A\|\|B\|$.
(a) Show that the exponential map $\exp : M(n, \mathbb{R}) \to \mathrm{GL}(n, \mathbb{R})$ is given by

$$\exp(A) = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots .$$

(you need to prove that the series is absolutely convergent)
(b) [Not required] Let $V = \{A \in M(n, \mathbb{R}) \mid \|I_n - A\| < 1\}$. Define

$$\log : V \to M(n, \mathbb{R}), \quad \log A = -\sum_{k=1}^{\infty} \frac{1}{k}(I_n - X)^k.$$

Prove: log is a smooth map form V to $\mathrm{GL}(n, \mathbb{R})$ and is the local inverse of exp.
(c) [Not required] Prove: For any $A, B \in M(n, \mathbb{R})$,

$$e^{A+B} = \lim_{k\to\infty} (e^{A/k} e^{B/k})^k$$
$$e^{[A,B]} = \lim_{k\to\infty} (e^{A/k} e^{B/k} e^{-A/k} e^{-B/k})^k.$$

(3) [**Lie group structure on S^3**]
Identify S^3 with the set of quaternions of norm 1.
(a) With the help of the multiplication of quaternions, show that S^3 is a Lie group, and determine its Lie algebra.
(b) (Not required) Show that as Lie groups, S^3 is isomorphic to

$$SU(2) = \left\{ \begin{pmatrix} z & w \\ -\bar{w} & \bar{z} \end{pmatrix} \mid z, w \in \mathbb{C}, |z|^2 + |w|^2 = 1 \right\}.$$

(4) [**Abelian Lie groups**]
(a) Let G be a Lie group. Prove: G is abelian (i.e. $g_1 g_2 = g_2 g_1$ for all $g_1, g_2 \in G$) if and only if $\mathfrak{g}$ is abelian (i.e. $[X_1, X_2] = 0$ for all $X_1, X_2 \in \mathfrak{g}$).
(b) (Not required) Prove: Any connected abelian Lie group of dimension n is isomorphic to $\mathbb{R}^r \times \mathbb{T}^{n-r}$ for some r.

(5) [**Simply connected Lie groups**] [Not required]
Show that the universal covering space of a connected Lie group admits a Lie group structure such that the covering map is a Lie group homomorphism.

(6) [**Surjectivity of the exponential map**]
(a) Show that $\exp : \mathfrak{so}(2) \to \mathrm{SO}(2)$ is surjective, and thus $\mathrm{SO}(2)$ is connected.
(b) Show that $\exp : \mathfrak{sl}(2,\mathbb{R}) \to \mathrm{SL}(2,\mathbb{R})$ is not surjective, but $\mathrm{SL}(2,\mathbb{R})$ is connected. [Hint: Show that $\mathrm{tr}(A^2) \geq -2$ for $A \in \mathrm{SL}(2,\mathbb{R})$.]

(7) [**The affine group**]
Consider the smooth manifold $G = \mathrm{GL}(n,\mathbb{R}) \times \mathbb{R}^n$. Define a multiplication operation on G via

$$(X, x) \cdot (Y, y) := (XY, Xy + x).$$

(a) Prove: G is a Lie group with respect to this multiplication.(the *affine group* of $\mathbb{R}^n$)
(b) Consider the vector space $\mathfrak{g} = \mathfrak{gl}(n,\mathbb{R}) \times \mathbb{R}^n$. Define a bracket on $\mathfrak{g}$ via

$$[(A, a), (B, b)] := (AB - BA, Ab - Ba).$$

Prove: $(\mathfrak{g}, [\cdot,\cdot])$ is a Lie algebra.
(c) Define a map $e : \mathfrak{g} \to G$ by

$$e(A, a) := \left(e^A, \sum_{n=1}^{\infty} \frac{1}{n!} A^{n-1} a \right).$$

For any $(A, a) \in \mathfrak{g}$, let $h(t) := e(tA, ta)$. Prove: $\{h(t) \mid t \in \mathbb{R}\}$ is a one-parameter subgroup of G.
(d) [Not required] Find a subgroup of $\mathrm{GL}(n+1,\mathbb{R})$ that is isomorphic to G, and explain how do we get $\mathfrak{g}$ and e above.

(8) [**Lie algebras of more linear Lie groups**]
For any $n \times n$ matrix B, we define

$$\mathrm{GL}_B(n,\mathbb{R}) = \{X \in \mathrm{GL}(n,\mathbb{R}) \mid X^T B X = B\}.$$

(a) Prove: $\mathrm{GL}_B(n,\mathbb{R})$ is a Lie subgroup of $\mathrm{GL}(n,\mathbb{R})$.
(b) Find the Lie algebra $\mathfrak{gl}_B(n,\mathbb{R})$ of $\mathrm{GL}_B(n,\mathbb{R})$.

(9) [**No small subgroup**] [Not required]
Show that any Lie group G does not have small subgroup, i.e. prove that there exists an open neighborhood U of $e \in G$ such that $\{e\}$ is the only subgroup of G that is entirely contained in U.